Given,

3- Layer fully Connected Network

with Dimensions [784 → 256 → 128 → 10]

(1) No. of. MAC's for each layer:

No. of. MAC's per layer = No. of. Neurons in the layer × No. of. inputs for each neuron in the layer.

No. of. MAC's for 1^{st} layer = 256 × 784

= 200704

No. of. MAC's for 2^{nd} layer = 128 × 256 = 32768

No. of. MAC's for 3^{rd} layer = 10 × 128 = 1280

(2) Total MAC's for One - forward pass :-

Total MAC's for 1 - forward pass = Sum of MAC's in all layers

= 1^{st} layer + 2^{nd} layer + 3^{rd} layer

= 200704 + 32768 + 1280

= 234752

(3) Total no. of. Trainable parameters [Weights only, No Bias]

Total No. of. Trainable Parameters = Sum of all layers (inputs × outputs)

= Same as MAC's as no-bias

= 234,752 weights

(4) Total Weight memory in Bytes [FP32] :-

In FP32, each Weight = 4 bytes

= Total parameters × Bytes/Weight

= 234,752 × 4 = 9,39,008 bytes

(5) Total Activation memory = Total no. of. neurons in network × 4

= [784 + 256 + 128 + 10] × 4

= 4712 Bytes

(6)

$$\text{Arthmetic Intensity} = \frac{2 \times \text{total MAC's}}{(\text{Weight bytes} + \text{Activation bytes})}$$

$$= \frac{2 \times 234752}{[939008 + 4712]}$$

$$= 0.4975 \text{ flop / byte}$$